TENT APPLICATION PUBLICATION
DIA
Date of filing of Application :13/11/2025

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B

(43) Publication Date : 19/12/2025

) Title of the invention : AEROVISION:- UAV-BASED AERIAL SURVEY AND VISION-BASED DATA AQUISITION

(51) International classification	:B60K 37/00, H04L 41/5067, B60K 37/20, B60K 20/08, H04L 41/507	**(71)Name of Applicant :** **1)Shrijeet Pagrut** Address of Applicant :CSE Department, SSGMCE, Shegaon, Maharashtra, India 444203 Maharashtra India **2)Atharva Ajay Raut** **(72)Name of Inventor :** **1)Shrijeet Pagrut** **2)Atharva Ajay Raut** **3)Devanshu Mankar** **4)Gargi Banarase** **5)Gauri Mankar** **6)Mr D. L. Bhombe** **7)Dr. D. D. Nawgaje**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
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(87) International Publication No	: NA	
(61) Patent of Addition to Application Number	:NA	
Filing Date	:NA	
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(57) Abstract :
The invention describes an integrated semi-manual Unmanned Aerial System (UAS) titled "Aerovision," combining an STM32-based multi-sensor circuit and onboard YOLO-based human detection. The system performs autonomous data acquisition and transmits synchronised results to a custom Aerovision web dashboard for ualisation. It includes environmental sensing, image processing, power management, and communication modules. The dashboard displays live detection, telemetry, environmental parameters, providing a unified interface for disaster response, monitoring, and analytics.

No. of Pages : 14 No. of Claims : 10

PATENT APPLICATION PUBLICATION

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) INDIA

) Date of filing of Application :04/06/2024

(43) Publication Date : 19/07/2024

4) Title of the invention : A DATA ANALYTICS CONFIGURATION FOR AN UNMANNED AERIAL VEHICLE

(51) International classification :B64C0039020000, G08G0005000000, H04B0007185000, G05D0001100000, H04N0007180000

(86) International Application No :NA
Filing Date :NA

(87) International Publication No : NA

(61) Patent of Addition to Application Number :NA
Filing Date :NA

(62) Divisional to Application Number :NA
Filing Date :NA

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(57) Abstract :
ABSTRACT A DATA ANALYTICS CONFIGURATION FOR AN UNMANNED AERIAL VEHICLE A controller module (100) for an unmanned aerial vehicle (109) comprising a sensing module (A) configure with said controller module (100) and an input module (B) configure with said sensing module (A) and said unmanned aerial vehicle (109). Further, a processing module (B) configure with said input module (C) and a display (116) can be configure with said processing module (B). However, said sensing module (A) senses the real time condition at particular environment and location in a manner said unmanned aerial vehicle (109) can fly and reach to the particular location and collect an input from an input module (C) in the real time condition of the particular location for said processing module (B) in order to process said input signal (114) to depicts the output on a display (116).

No. of Pages : 14 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :11/12/2025

(21) Application No.202521125232 A

(43) Publication Date : 23/01/2026

(54) Title of the invention : RESCUE ALERT SYSTEM: HUMAN SAFETY DEVICE USING WEARABLE PCB AND IOT

(51) International classification	:H04W 4/90, G08B 25/01, G08B 25/10, H04M 11/04, G08B 21/02	**(71)Name of Applicant :** **1)Shrijeet Pagrut** Address of Applicant :CSE Department, SSGMCE, Shegaon, Maharashtra, India 444203 Maharashtra India **2)Nivrutti Dattatray Raut** **(72)Name of Inventor :** **1)Nivrutti Dattatray Raut** **2)Parth Kailasrao Deshmukh** **3)Sudhir R. Paraskar** **4)Saurabh S. Jadhao** **5)Shrijeet B. Pagrut**
(31) Priority Document No	:NA	
(32) Priority Date	:NA	
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Filing Date	:01/01/1900	
(87) International Publication No	: NA	
(61) Patent of Addition to Application Number	:NA	
Filing Date	:NA	
(62) Divisional to Application Number	:NA	
Filing Date	:NA	

(57) Abstract :

The Rescue Alert System is an IoT-based safety device designed to provide instant help during emergencies such as kidnapping, harassment or accidents. In many cases, victims cannot make a call or share their location in time, which delays rescue operations. To address this problem, the proposed system uses GSM and GPS technologies to automatically send an SOS message containing the victim's live location to a registered mobile number when the emergency button is pressed. The system integrates components such as the Arduino microcontroller, SIMA7670C GSM module and NEO-M8N GPS module, assembled on a flexible PCB to make the design compact, portable and Wearable. Testing results show that the system delivers quick response and stable communication with low power consumption. This innovation provides a reliable, cost-effective and user-friendly solution for real-time rescue operations. It can significantly reduce response time and improve safety for women, children and elderly individuals. The project demonstrates how technology can play a vital role in enhancing personal security and reducing unsolved emergency cases through smart, real-time communication.

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(21) Application No.202421043408 A

(43) Publication Date : 19/07/2024

(54) Title of the invention : A DATA ANALYTICS CONFIGURATION FOR AN UNMANNED AERIAL VEHICLE

(51) International classification :B64C0039020000, G08G0005000000, H04B0007185000, G05D0001100000, H04N0007180000
(86) International Application No :NA
Filing Date :NA
(87) International Publication No : NA
(61) Patent of Addition to Application Number :NA
Filing Date :NA
62) Divisional to Application Number :NA
Filing Date :NA

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ABSTRACT A DATA ANALYTICS CONFIGURATION FOR AN UNMANNED AERIAL VEHICLE A controller module (100) for an unmanned aerial vehicle (109) comprising a sensing module (A) configure with said controller module (100) and an input module (B) configure with said sensing module (A) and said unmanned aerial vehicle (109). Further, a processing module (B) configure with said input module (C) and a display (116) can be configure with said processing module (B). However, said sensing module (A) senses the real time condition at particular environment and location in a manner said unmanned aerial vehicle (109) can fly and reach to the particular location and collect an input from an input module (C) in the real time condition of the particular location for said processing module (B) in order to process said input signal (114) to depicts the output on a display (116).

No. of Pages : 14 No. of Claims : 10

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(22) Date of filing of Application :05/07/2025

(21) Application No.202523064402 A

(43) Publication Date : 08/08/2025

(54) Title of the invention : SYSTEM FOR MAPPING OF FLOOD AND EARLY WARNING CONFIGURATION BY MEANS OF AN UAV

(51) International classification	:G06F0018232130, G06Q0050260000, G08B0021100000, G06Q0020100000, H04Q0009000000
(86) International Application No	:NA
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(87) International Publication No	: NA
(61) Patent of Addition to Application Number	:202421043408
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(62) Divisional to Application Number	:NA
Filing Date	:NA

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(57) Abstract :

ABSTRACT The present invention relates to an unmanned aerial vehicle(109) (UAV) system equipped with a controller(108) module for real-time flood monitoring, analysis, and disaster response. The system comprises an input module(C) for collecting environmental and UAV-specific data (110b) using specialised sensors, a processing module(B) powered by a microcontroller to perform spatial analysis, predictive analytics, and 3D flood mapping(121), and an output module (D) that delivers actionable insights through a web interface(119) and display unit(115). The system integrates tools such as ArcGIS and WebODM for terrain modeling and employs artificial intelligence techniques—including CNNs for image segmentation, Random Forest/XGBoost for flood prediction, LSTM for level forecasting, and K-Means clustering for risk classification. Additionally, the system provides real-time visualisation, early warning alerts, post-disaster damage assessment, and supports flood prevention and evacuation planning. This invention offers a comprehensive, scalable, and adaptive UAV-based solution for effective flood risk assessment and management.

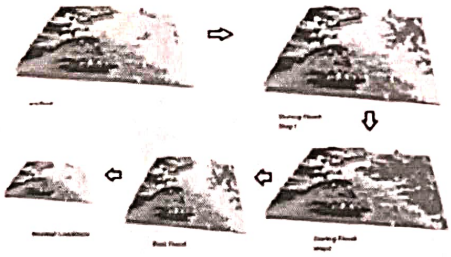

Figure no. 02

No. of Pages : 13 No. of Claims : 10